25CSA502A	OBJECT ORIENTED PROGRAMMING USING JAVA	L-T-P-C: 3-0-2-4

Course Objectives

- The main objective of this course is to understand the basic concepts and techniques that form the object-oriented programming paradigm using the Java Language.

Course Outcomes

COs	Description
CO1	Identify classes, objects, members of a class and relationships among them needed for a specific problem.
CO2	Implement Java application programs using OOP principles and proper program structuring.
CO3	Demonstrate the concepts of polymorphism, inheritance and thread and document a Java Program using Javadoc.
CO4	Use Java AWT and Swing classes to build GUIs and understand how collection interface is implemented.
CO5	Demonstrate the Conceptual model of UML, activity diagram and their modelling techniques.

CO-PO Mapping

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	3	2	-	2	1	-	-	1
CO2	3	3	-	3	1	-	-	1
CO3	3	2	-	2	1	-	-	1
CO4	3	3	-	1	1	-	-	1
CO5	3	3	-	3	-	-	-	1

3-strong, 2-moderate, 1-weak

Syllabus

Unit I
Introduction to object-oriented software design, Comparison of programming methodologies, Object Basics, Java Environment, Classes and Objects, Data Members, Access Specifiers, Arrays within a Class, Array of Objects, Constructors, Default Constructors, Destructors, Static Members, Constant Members.

Unit II
Overview of Streams, Bytes vs. Characters, File Object, Binary Input and Output, Reading and Writing Objects, Method Overriding, Polymorphism, Super, Interfaces and Abstract Classes, Packages, Exception

Unit III
Introduction to Threads, Creating Threads, Thread States, Runnable Threads, Coordinating Threads, Interrupting Threads, Runnable Interface, Synchronization.

Unit IV
Collection framework, Collection interfaces and classes, AWT, Swing, Event Handling, Javadoc

Unit V
Object Oriented Design with UML, Class, object diagrams and sequence diagrams. Use case diagrams and activity diagrams

Textbooks / References:

1. Herbert Scheldt, “Java: The Complete Reference, Eleventh Edition”, Oracle, 13th edition, 2024
2. Deitel PJ. Java how to program. Eleventh Edition, Pearson; 12th Edition, 2025.
3. Nino J, Hosch FA. Introduction to Programming and Object-oriented Design using Java. Wiley India Private Limited; 2010.
4. Naughton P. and Schildt H. Java 2: The Complete Reference. Eighth Edition, Tata McGraw- Hill; 2011.
5. Bahrami A. Object Oriented Systems Development. Second Edition, McGraw-Hill; 2008.
Booch G, Maksimchuk RA. Object-oriented Analysis and Design with Applications. Third Edition, Pearson Education; 2009.**AMRITA VISHWA VIDYAPEETHAM**

Amrita Online - Course Plan

Week	**Topic**	**Learning Objectives**	**eLearning Content**	**Video Lectures**	**Assessments**
1	Introduction to OOPs concepts	Get an Overview of Object-Oriented Programming Concepts	6 videos	Introduction to Object Oriented Programming Encapsulation and Abstraction Class and Object - An Intro Polymorphism Inheritance in Java Summary of OOPs	Quiz 1
2	OOPs in Java	To study class and constructors in Java	6 videos	Java Evolution Features of Java Java Source File Lab Demo - Eclipse IDE Lab Demo - Declaring Variables Lab Demo - Reading User Input	Quiz 2
3	Java Basics-I	To study how to write programs in Java	10 videos	Data types Operators Selection Statements Loops in Java Jump Statements Types - Demo Lab Demo - Selection Lab Demo - Loops1 Selection StatementsI Datatypes and VariablesLabdemo	Labsheet 1 Quiz 3
4	Java Basics-II	To study how to write programs in Java	11 videos	Classes Intro Constructors Access Specifier Introduction to Arrays String Class StringBuffer Class Lab Demo Loops2	Quiz 4 Labsheet 2

				Lab Demo Switch case	
				Lab Demo Jump Statements	
				Typecasting Labdemo	
				Arrays lab demo	
5	Polymorphism	To understand polymorphism	8 videos	Intro to Polymorphism	Labsheet 3 Quiz 5
				Static Binding	
				Dynamic binding	
				Constructor Overloading	
				Overloading Vs Overriding	
				Lab demo polymorphism	
				Constructor lab demo	
				Lab demo-strings	
6	Inheritance	To understand inheritance	8 videos	Intro to Inheritance	Quiz 6 Labsheet 4
				Inheritance Types	
				Multi level Hierarchy	
				super	
				inner class	
				Inheritance I	
				Inheritance II	
				Inheritance III	

7	Details of Interface and Implementing methods	Get an idea about how to declare interfaces. Introduce packages, built in packages and user defined packages	9 videos	Abstract class	Labsheet 5 Quiz 7
				interface	
				interface example	
				package I	
				package II	
				package III	
				Lab Demo Setter and Getter	
				Lab Demo Static	
				lab demo package	
8	Exceptions	Error and Exception handling mechanism in Java using try, catch, throw	8 videos	Exception Intro	Quiz 4 Quiz 8
				Exception Type	
				Built-in Exception	
				Throw Exception	
				User defined Exception	
				exception handling demo I	
				exception handling demo II	

				exception handling demo III	
9	Streams	Perceive concept input and output streams	3 videos	Streams Intro Console Reading & Writing File Read & Write	Quiz 9
10	Thread	To understand basics of threading	5 videos	Thread vs Process Thread creation Thread priorities Thread Synchronization Multithreading demo-1 Multithreading demo-2 Multithreading demo-3	Quiz 10
11	GUI	To design GUI applications using Java.	5 videos	GUI intro Swing intro Swing Hierarchy Event Handling Demo on Swings and Event	Labsheet 6 Quiz 11
12	Generic	To define our own classes with generics type which is a class or interface that is parameterized overtypes	5 videos	Generics I Generics II Generics III Generics Restrictions Generics Hierarchy	Quiz 12
13	Introduction to JDBC	To understand the basic concepts of JDBC and to understand the basic CRUD operations using JDBC	5 videos	JDBC introduction Steps in JDBC JDBC Statements CRUD JDBC Demo	Labsheet 7 Quiz 13
14	Unified Modeling Language	To understand the basic concepts of Unified Modeling Language	4 videos	UML introduction UML class diagram UML relationships Usecase diagrams	Quiz 7 Quiz 14

Evaluation Plan

Internal	**14 Quizes(best 7 will be taken)**	**30**	**100**
External	• **MCQ-50** • **Descriptive-10** • **Practical-10**	**70**	

7 mandadory assignments will be given which are nongraded

Faculty Information

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